IGVC Rundown

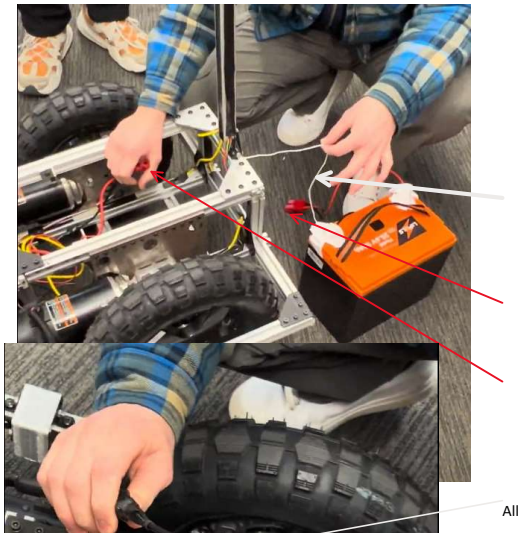
Tuesday, January 20, 2026 11:20 AM

Written by: Jerry E Trejo

This document was created at the beginning of the 2nd semester for the IGVC spring 2026 team. Though somethings have changed during the semester the basis of the section below is still fundamental to the understanding of the entire IGVC vehicle hardware architecture. This document is split into two sections. The first section (the below one) is the section as stated before was one I made to understand the entire hardware architecture the previous team had created. The 2nd section which is the one below the first is a summary of the things that we have changed or added which isn't in full depth as the first section explains all if not most of the technical and basics of it all that you need. So it is advised to go through both sections.

- The cameras work good individually but not together as they didn't have enough time to test them to work together
- The lidar and camera work good individually but don't work good together as didn't have enough time to test
- The lidar or cameras start with a detection of the left side of an object. It start with a priority of the left lane. Its follow the left lane until the line disappears or there is an object in the way.
  - The code does start on the left side of the lane of a two lined lane.
  - We will have to set the rule depending on the IGVC competition on where does the lidar start from like where in the lane left side right side middle of the lane. Where does it start from?
  - The way the code is set up its to scan the left side if something is in front of it then it switches to scanning the right side (here they may be talking about changing to the right side camera to check and see if the right side is free.) If the right side is seen to be free then the car will turn to the right until it gets out of the way of the object and continue riding on the right side. The car checks ft ahead to change directions before even arriving to the object.
    - If something is on both lines of the lane and only has room in the middle I think they were saying that it checks both sides continuously checking to see if both objects give the same distance of leeway on each side to go in between them. It keeps checking both sides all the way pass the objects.
      Though not sure if they implemented this already or if they said that is what we should implement for the test requirements.
- He said that in the IGVC competition there is rules against using gps to guide you but not sure to check the rule book for that. (Not sure what he means here as in not using gps for things like camera and lidar demonstration? Or at all in general)
- **Integrating the camera the LIDAR and the GPS is the main thing we need to do.**
  - We also have the encoders installed (what are encoders?) so if you want to use the encoder information to keep the wheels going the same RPM to keep going straight.
  - IF we send PWM at 15% on both wheels the RPM on both wheels will be different as one of the wheels is faster.
    - They put 15% for one motor and 16% for another one to get the same RPM
  - They say that the PWM to RPM is pretty consistent. As in if they put 15% they would expect to get the same RPM every single time. But that also it depends on the environment as well. In the actual compition there will be a lot of slopes that will need to be climbed. Which will affect the alignment of both motors. So we need to figure out that problem how to calibrate the RPM to be the same no matter the environment.
  - They have encoders on both motors to get the RPMs
  - 5:22:05
    We have the PCB build up already so it has the NRF encoders you just hook up the wires there already placed in there. And you already have the code its just not combined

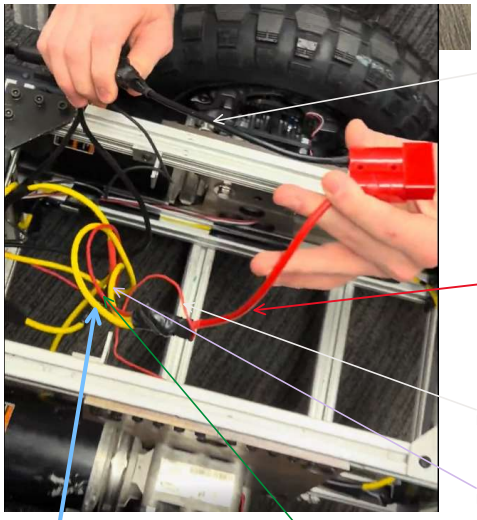
  - They say that the way they have it made is that the camera and the lidar are working together ^^or not?????
- Wide lens camera can be used for seeing the front and the lane of the side it is on.
  - You can use this one for identifying objects stop signs and other things getting in the way.
  - The code for one camera is not the same but is similar to the code for the other camera that is not wide lens.
    the narrow lens if programmed using the wide lens code would stretch the distance that was hard coded into to locate the lanes.
    Calibrated to be a specific distance from the lane.
  - The way lane detection works is that you have a green line on the screen and a red dot at the bottom of the lane we want the green line lined up with the red dot and then we calibrate it to the specific distance we want
    - So the car would move left or right to try to line up the red dot with the green line or vice versa. Where if to much left it would move to the right to be center again.
  - If you change camera you would need to change values, distance, points, saturation levels of lighting, etc.



Common ground for the lights

Connect these to turn on everything

All the ground for all the components and devices are here



All the ground for all the components and devices are here

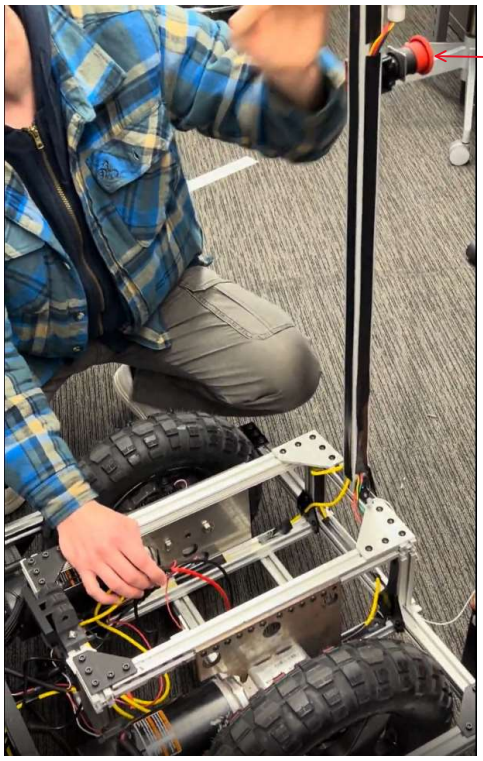
All the power for all the components and devices are here

Power for the lights

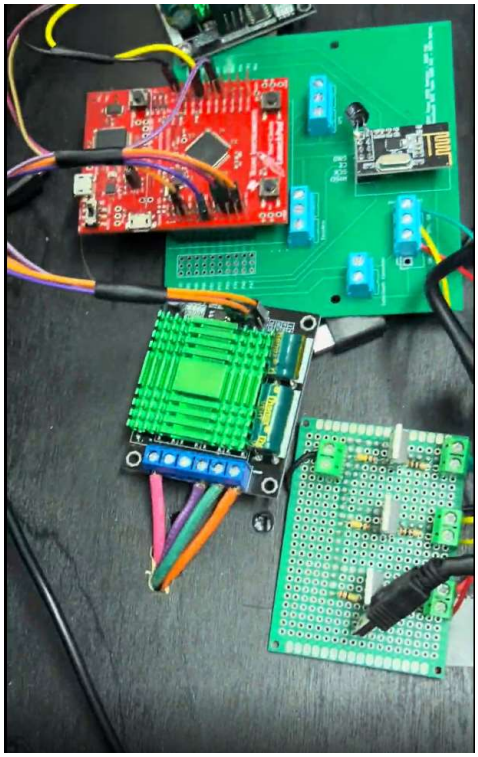
Power to the 5V power converter

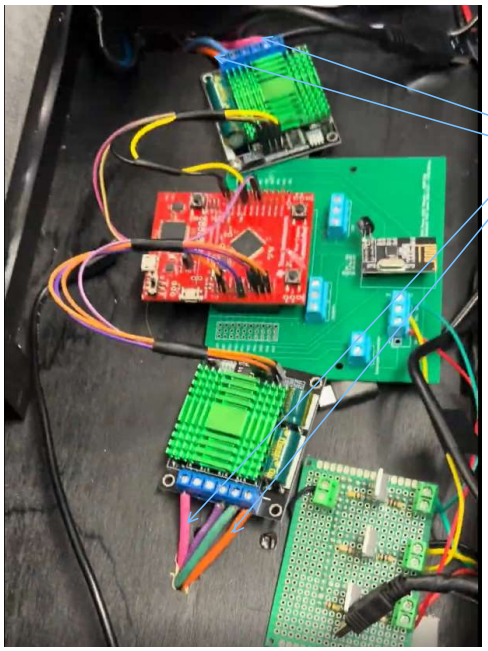
Power to the motors

Power to the 12V power converter

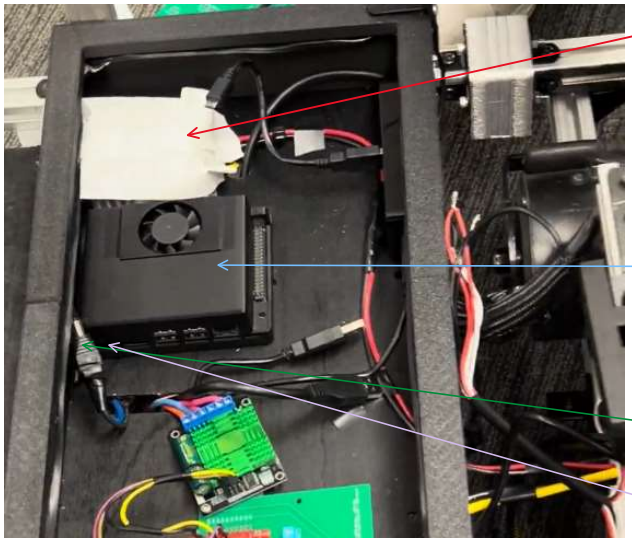


Emergency switch: Connected to the positive side and cuts off power to the motors. AND ONLY THE MOTORS!! Everything else is still on. If pressed turns off DCC to the motors.





The Emergency button is connected to the pink and orange cables on both boards.
And those cables are the 26V that the PWM uses to output to the motors.

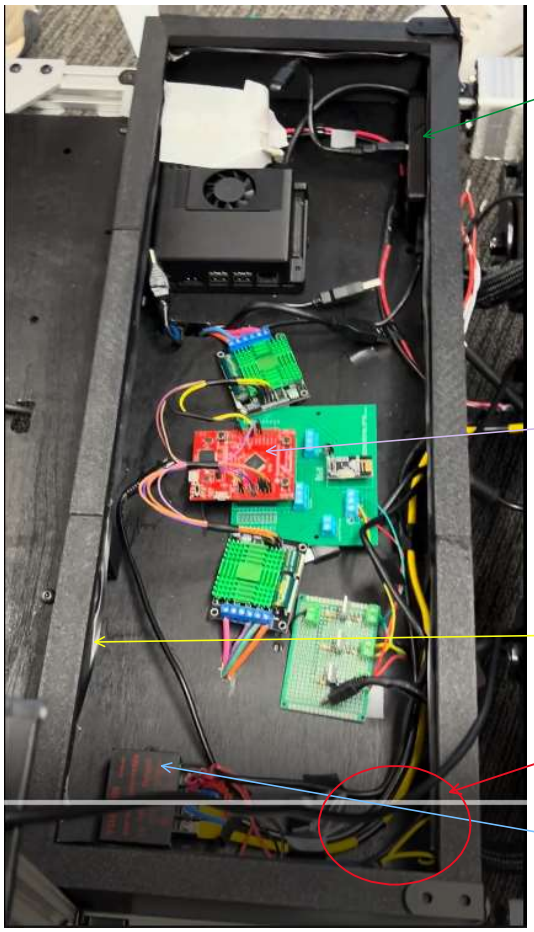


12V Power Converter: powers only the jetson nothing else is connected to it.

Jetson

12V Power Converter Cable to Power Jetson.

12V Power Converter Cable to Power Jetson is plug in here



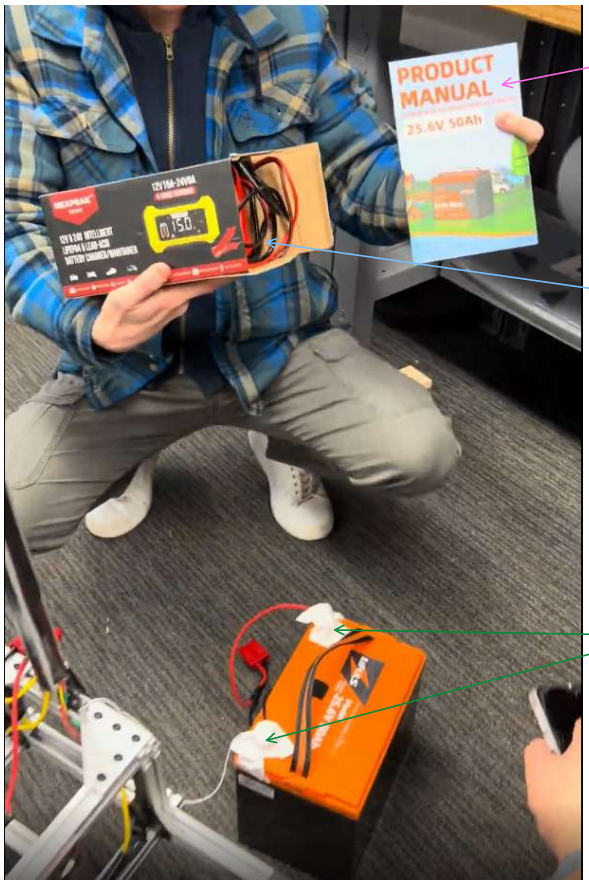
USB Hub

Tiva Microcontroller

LED RGB lights: Located all around the rim of the box. (AESTHETICS PURPOSES ONLY)

Arduino Microcontroller **(we don't have one we need to order one as it was Andy's)**: that controls the outputs of the lights

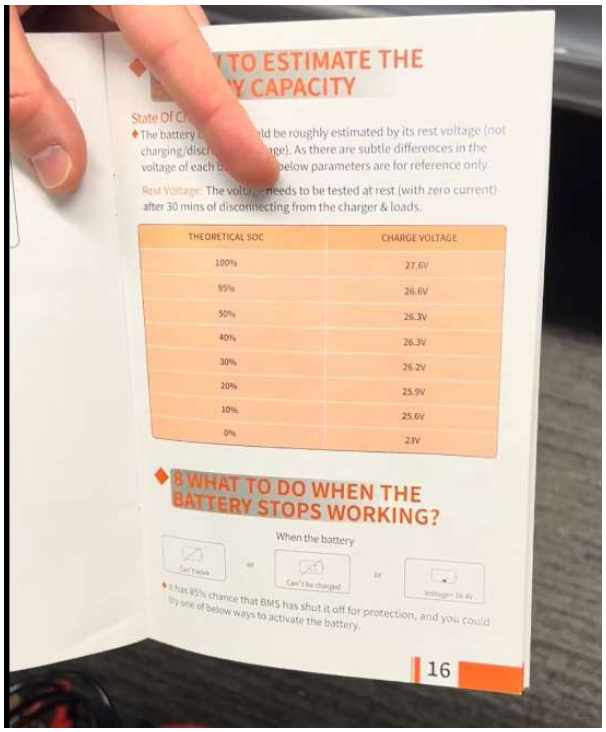
5V power converter



The manual for the battery

The Charger for the battery

The battery its self: There is no need to remove anything that is show right here unless you are charging it.
The tape has to go back on every time you take it off and should always be on. The Conly and Steve have already got on the last teams ass because they didn't do that.
SO KEEP TAPE ON AT ALL TIMES WHEN NOT CHARGING.
DON'T LEAVE THE ROOM WHEN CHARGING. Batteries can overcharge and explode.



TO ESTIMATE THE
Y CAPACITY

State Of C

The battery ld be roughly estimated by its rest voltage (not charging/disch ge). As there are subtle differences in the voltage of each b below parameters are for reference only.

Rest Voltage: The voltage needs to be tested at rest (with zero current) after 30 mins of disconnecting from the charger & loads.

THEORETICAL SOC	CHARGE VOLTAGE
100%	27.6V
95%	26.6V
50%	26.3V
40%	26.3V
30%	26.2V
20%	25.9V
10%	25.6V
0%	23V

8 WHAT TO DO WHEN THE BATTERY STOPS WORKING?

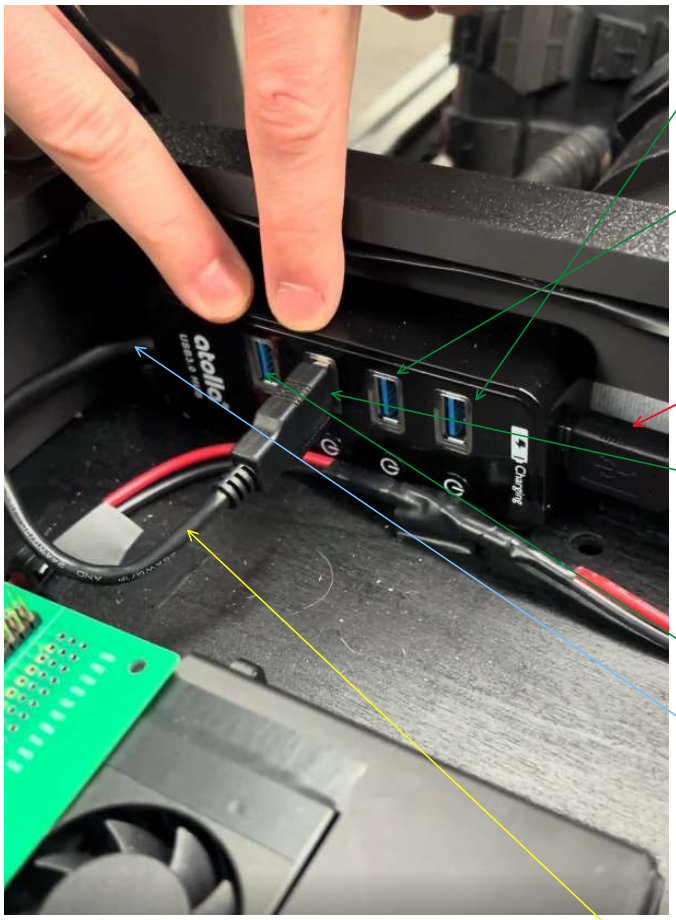
When the battery

Can't work or Can't be charged or Voltage< 18.4V

It has 85% chance that BMS has shut it off for protection, and you could try one of below ways to activate the battery.

16

Page 16 in the Battery manual tells you how much percent charge the battery is.
It is industry standard to not let batteries go under 20% of their charge as cells in the lithium battery may not be able to charge up.
Which decreases the batteries life if done multiple times.
Overcharging can lead to the battery exploding so DON’T LEAVE IT CHARGING ALONE. SOME ONE MUST ALWAYS BE CHECKING IT.



Camera

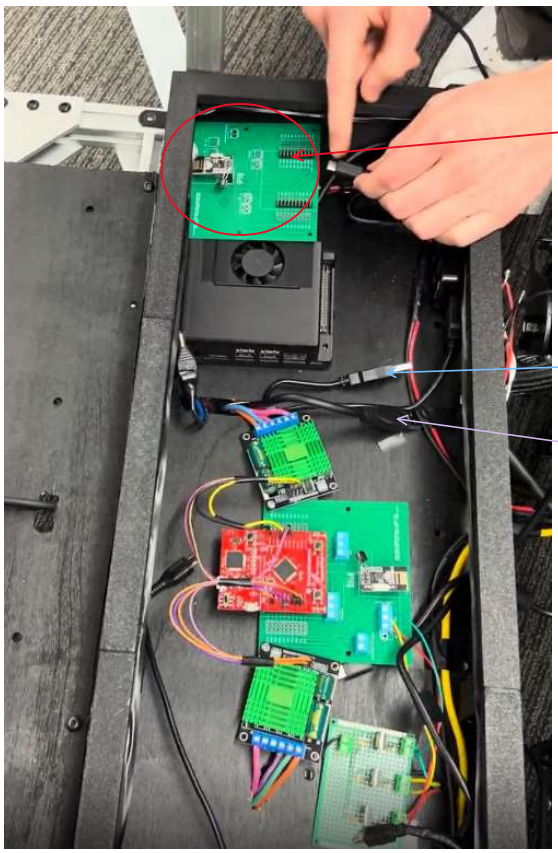
Camera

This cable supplies power to the USB Hub

Powers the Red Board (Tiva microcontroller)

Lidar

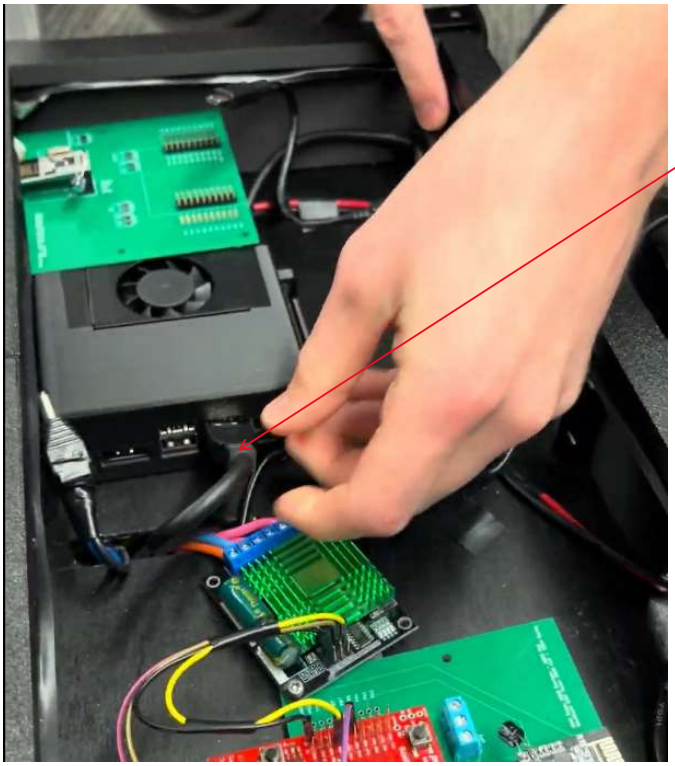
This is how the jetson gets all the information from the Lidar and Cameras.



Another red board goes right here. The green board is loose and not connected to anything so might need to look for it.
The cable in the hand is this to power the red board.

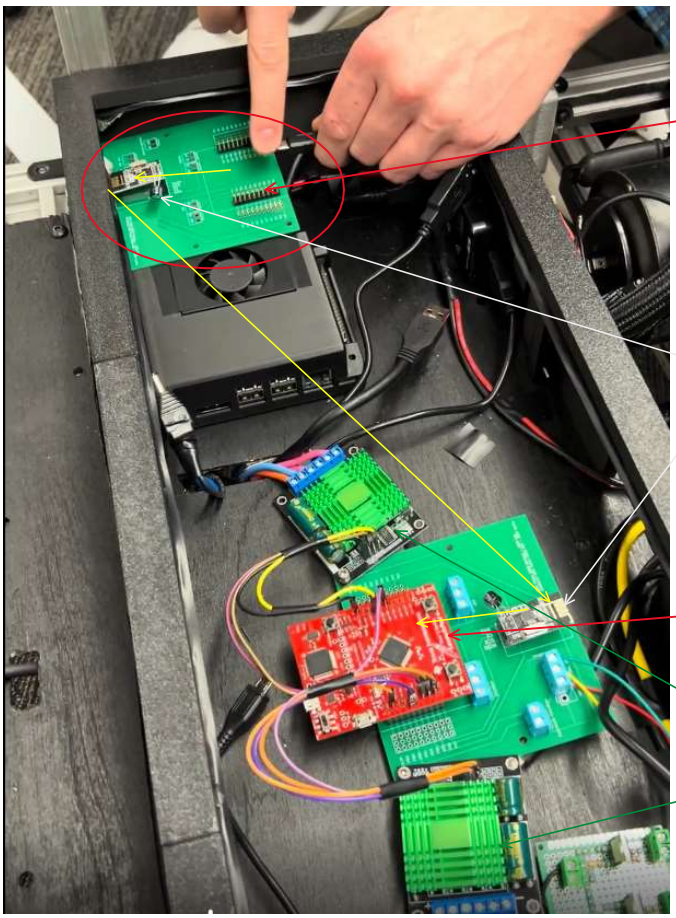
LIDAR Cable need to be connected to the USB HUB.
There is no particular order for how things should connect to the USB HUB but the way shown before is the same way it was presented to us.

USB Cable for the USB HUB to send data to the Jetson.



This USB cable is the cable for the USB Hub to send data to the Jetson. And it connects to that USB port its going in shown in the picture.

There is no particular port for the USB cable to go in but a past team had troubles with the other USB ports as they got damaged so if your having troubles with a port just change it.

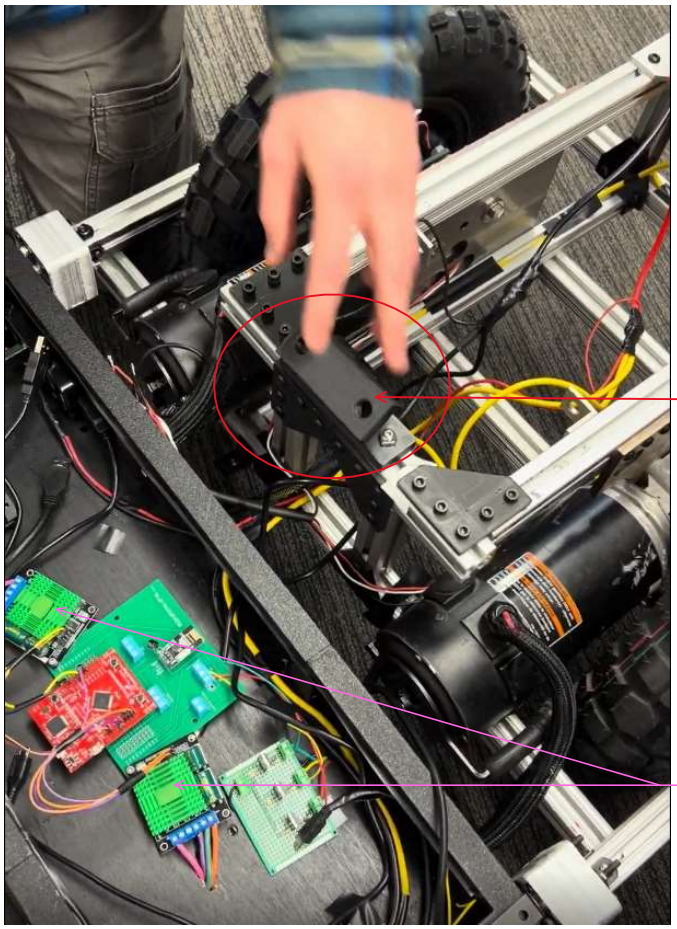


The way the jetson controls the motors. Is that it takes in information from the lidar, and cameras. Then it outputs the movement through the transmit red board. Over wirelessly through the receive red board.

NRF modules: there is a transmit command and a receive command. The transmit command has a motor powers that it supplies.

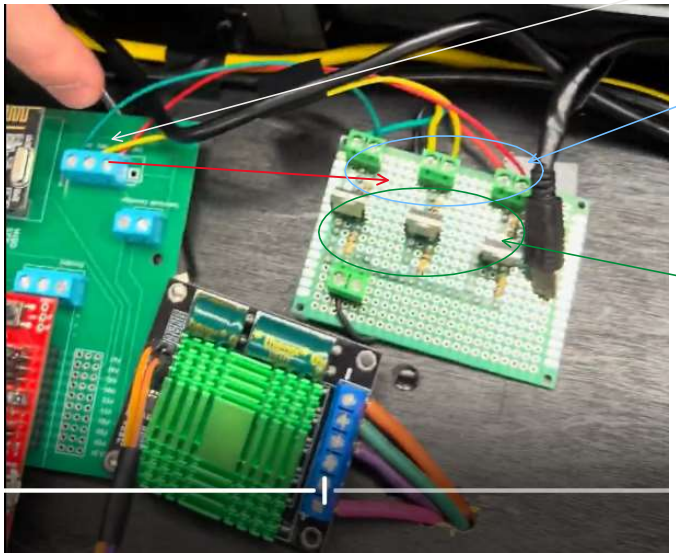
Once the receive red board gets the information, what ever command it sends, it outputs the PWMs

You don't want the jetson to have a direct connection to the motor drivers



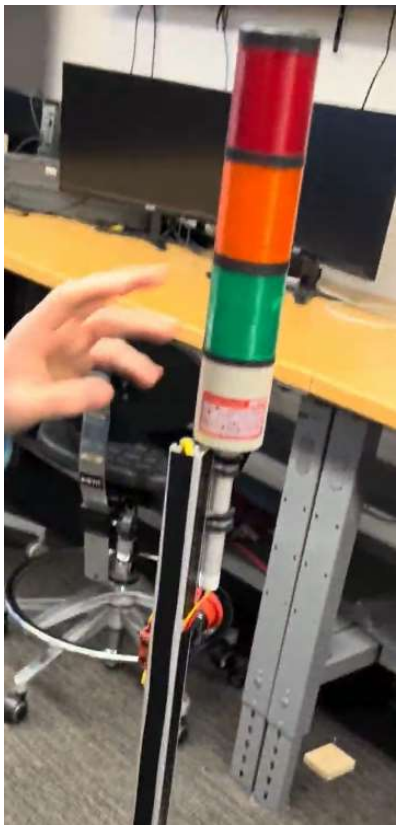
This is where the LCD Screen goes **We don't have one we need to order one**

When you get the command here forward, backward, turn left, turn right, what ever it is. It'll output the motor speeds and it'll also output the emergency lights through these three wires.

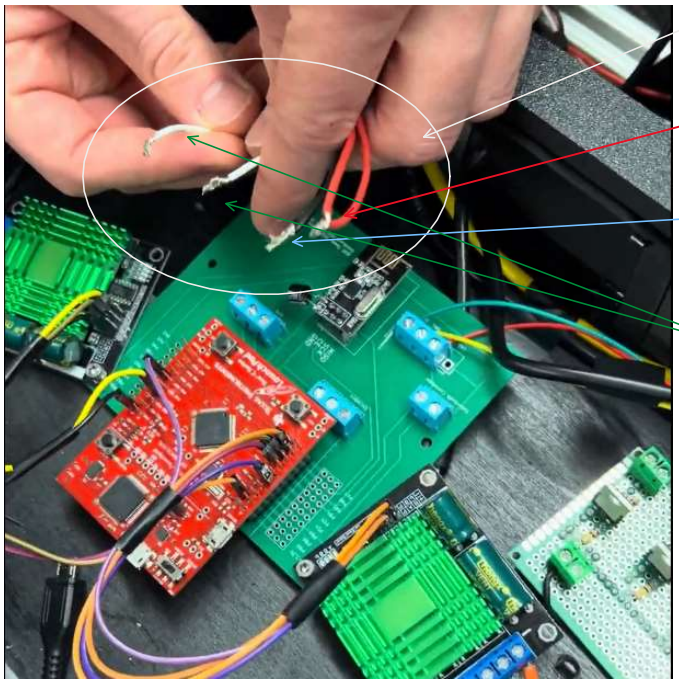


The three wires (green, red, and yellow) are pulled here

These three connections are used through GPIOs.
So whatever the gpios are set to it turns on the lights



There are specific patterns they have to display set in the rule book.
So we will have to mess with the GPIO pins to see which need to be turned on or off depending on the state that we are in. If we are in autonomous mode, if we are controlling it, or if we are stopped.



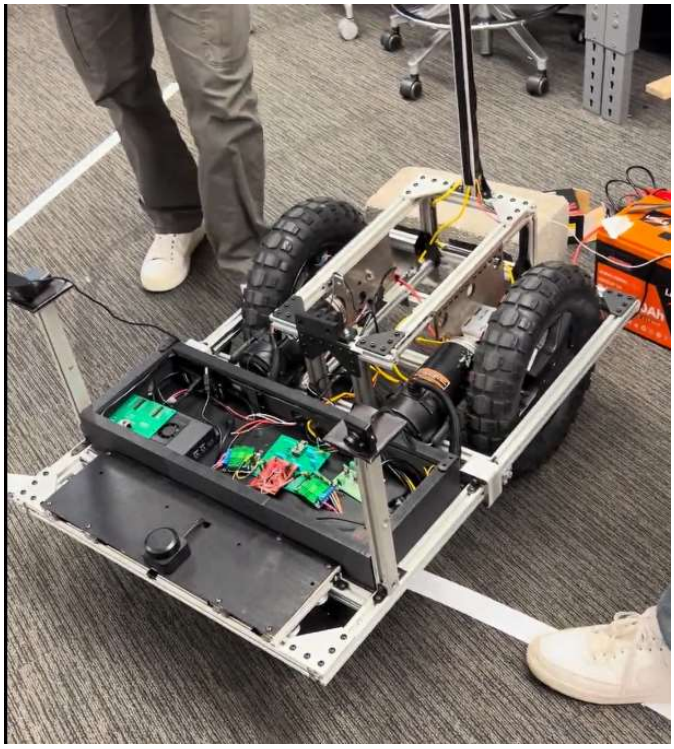
Wires for the encoders. To the two motors.

Power for the encoders

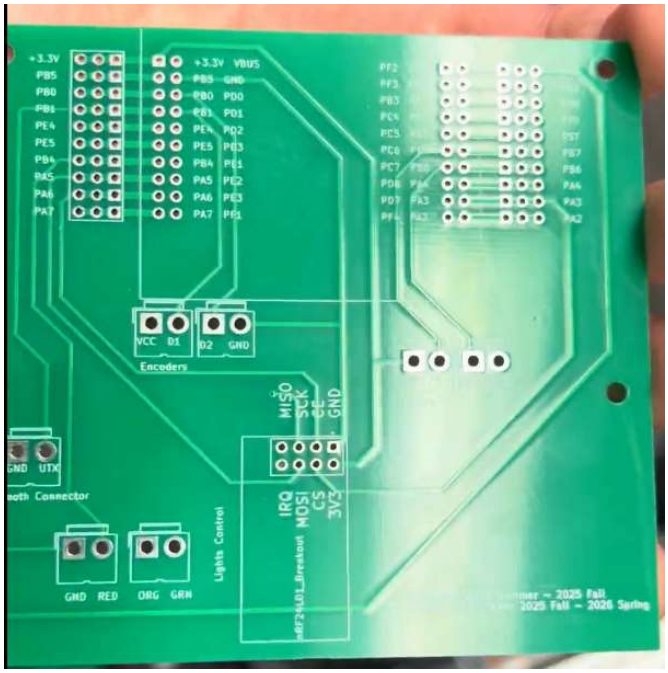
Ground for the encoders

The two encoder data lines. We will need to test out to see which line is for which motor. The left or right.

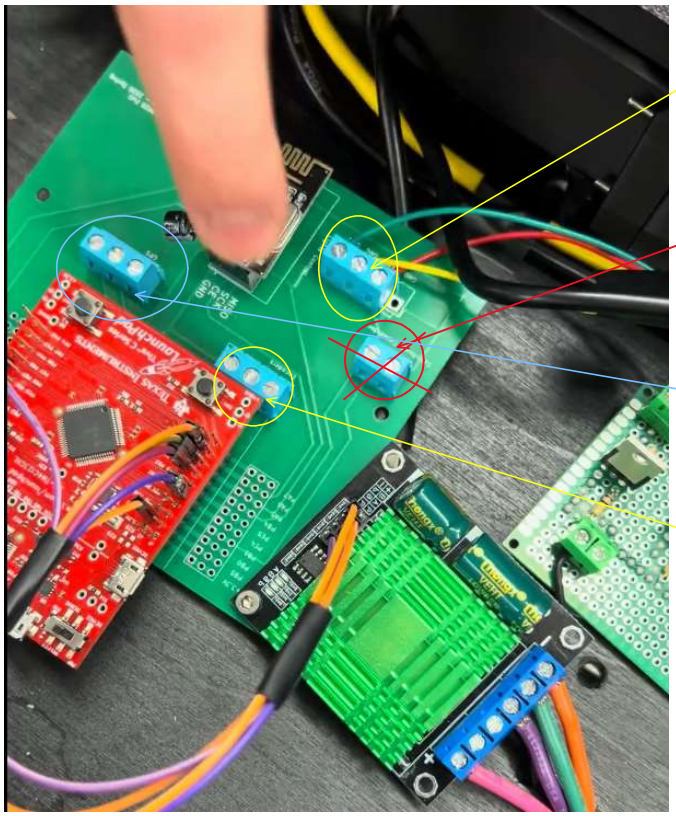
If you turn on one motor you can read the data lines to see which data line read data. It would show the revolutions per minute.



To test which encoder data line is for which motor.
You can use the cinder block to lift the back of the car up so that the wheels aren't touching the ground. Then turn on one of the wheels. Then read the encoder data lines and see which one is outputting data Revolutions/min.



This PCB has all the information on what the pins are. The ones already in use in the car have the labels covered. So use these boards as a guide to know what is what and connect everything properly. You can find these in the IGVC dedicated box.
This design is also on their GitHub as well.



For lights

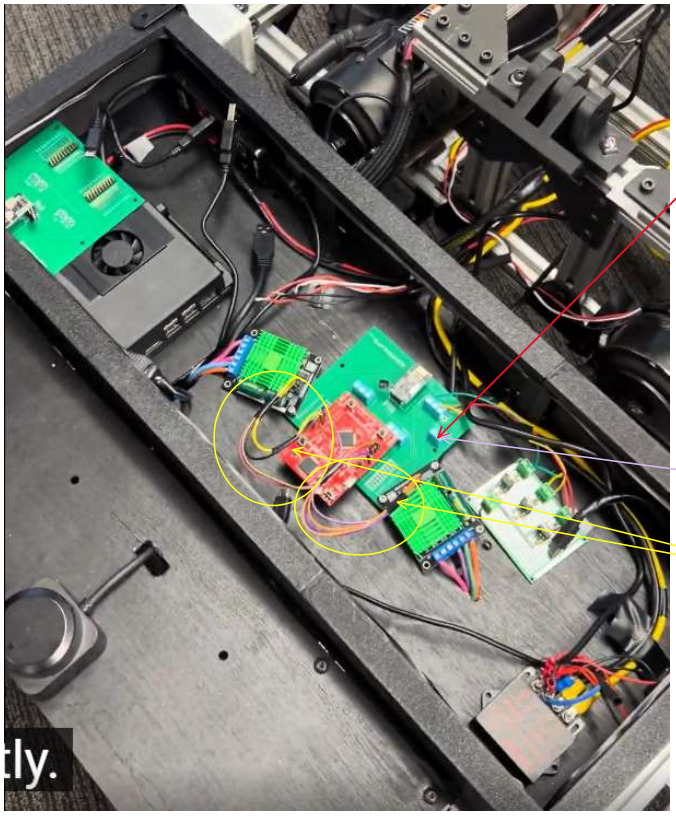
For sabretooth but isn't needed anymore as changed the motor drivers.

Once we get GPS plugged in it will go here.

For Encoders

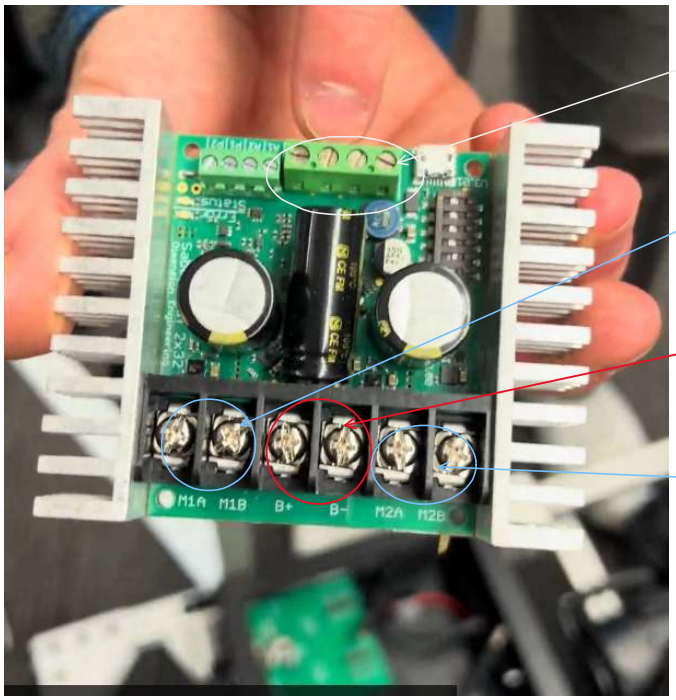
In summary: When you send a command from the 2nd red board over transmit it will receive it over the receiver. Receiving the command to the 1st red board (the one in picture). You can output through the lights. You can take the inputs from the encoders and take the input from the GPS. All which is done within the red board.

What Kopil suggests us to do is to buy a 2x32 sabertooth motor driver. And use serial lines pins in the sabertooth and not use the micro USB cable. And use the red board to connect to the sabertooth instead of using jetson.
The green motor driver that they ended up using are cheap chinease motor driver that have no data sheet, but they were able to get it to work. By going through the code for the motor drivers we can figure out how they work.
They work just like a regular motor driver. They have a PWM channel, some directions, and power. They were suppose to have two channels but only one channel works.
So it's a defective product that is why they had to use two different motor drivers. As only one channel works on both of them.
For now they work but they may not in the future. That's why its best to just buy and replace them with a nice 2x32 sabertooth motor driver.
Sabertooth is about 140$



Plus its super easy to set up the sabertooth to the current set up as the pins are already set to be used for the sabertooth.

All these cables and connection are all equivalent to this established connection for the sabertooth.



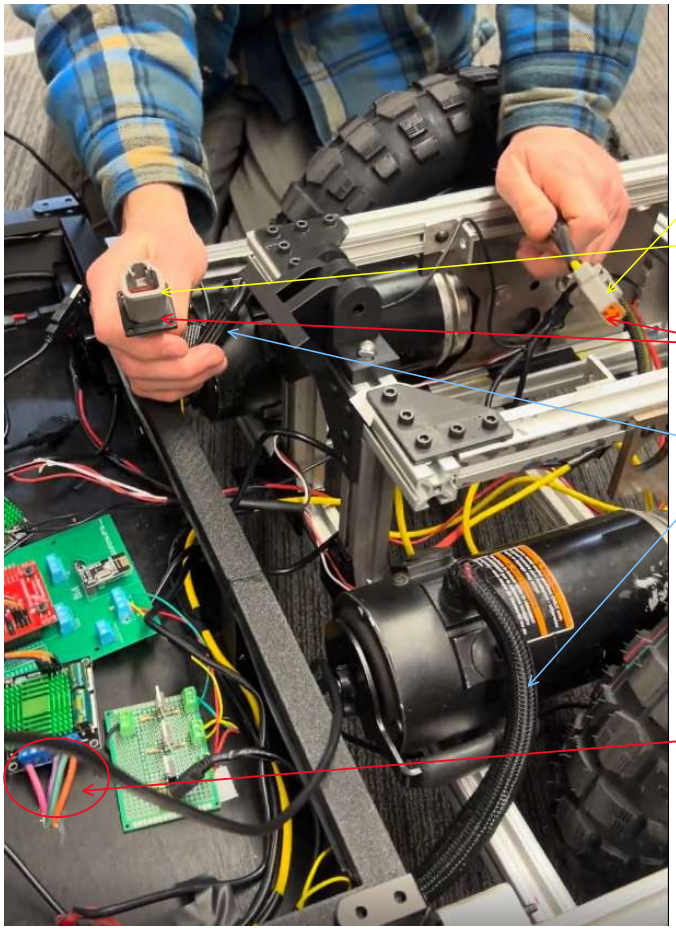
Serial lines

Motor 2

Main power

Motor 1

This is the sabertooth: For the sabertooth it is only one input pin you have to use compared to the current motor drivers that have a bunch of wires. On the sabertooth it is just two pins. As everything else is just done with UART.



How the Motors are connected

Male connection for the main power.

TO DO: BEFORE YOU UNPLUG THE 26V BATTERY
YOU SHOULD DISCONNECT THE MALE AND FEMALE CONNECTIONS.

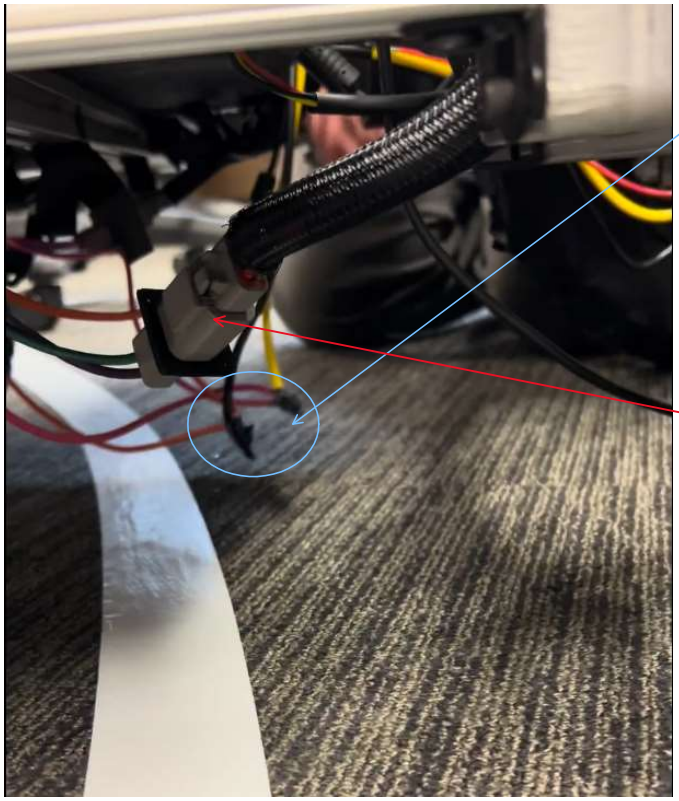
Female connection for the main power.

The Main power connections are output connections to the motor drivers.

Suggestion: The other team (kopal) advice us to change these headers as he says that they don't click and can maybe lose connection in the future. Though that hasn't happened to them so its just a suggestion.
I myself looked it over and see no problem with it. Its strong tight and holds well. But it's a suggestion nonetheless.

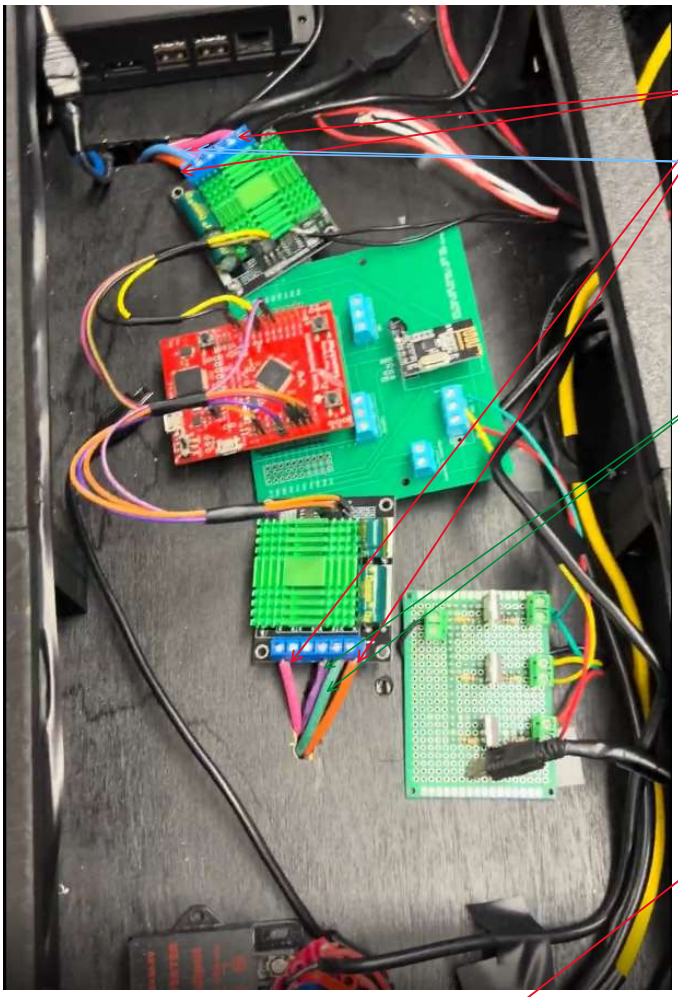
These cables are inputs for the motors and input connections for the motor drivers.

These cables are the inputs and outputs of the cables mentioned above.



These cables can be identified from under the car.
These are the cables for the main power output. The black and yellow cables carry 26V to each of their corresponding motor drivers.
The wires attached to those cables are the ones that lead to the motor drivers.
Those cables are both pink and orange on both motor drivers.

There are two cables like this one that are connections to the motors inputs.
This one is connected to green and purple cables.



These pink and orange outer wire connections are connected to main power output female cable. And are inputs to the motor drivers.

Here the inner pink and blue wires are the connected to the motor inputs to the right motor and are outputs to the motor driver.

The motor drivers get the input from the 26V (pink and orange wires). It then gets the signal for the PWM and the direction pins (the multiple taped wires connected to each motor driver). And then it outputs from the green and purple or the blue and inner pink.

The purple and green wires here are connected to the motor inputs to the left motor and are outputs to the motor driver.

FINAL WORDS OF WARNING:
If the USB Hub is getting power through the 5V converter, then the blue lights on the USB HUB are on. And if the USB Hub is still connected to the Jetson then the it's sending power to the Jetson. This means if you unplug the power to the Jetson, but the USB Hub is still connected to the Jetson then the USB ports on the Jetson could get fried. You will hear a loud high pitched squeal that is coming from the jetson USB ports. This is bad.

DISCONNECT THE USB HUB POWER BEFORE TURNING OFF THE JETSON!!! OR BETTER YET DISCONNECT THE USB CONNECTION FROM THE USB HUB TO THE JETSON THAT WAY YOU DON'T HAVE TO TURN OFF THE POWER FROM THE USB HUB ALL THE TIME. THEN TURN OFF THE JETSON POWER.

Jetson Password is: secure

The tool gun thing is IGVC's

Right now the way that the Car is set up with the chinease motor drivers there is no current way the past team knew how to implement the wireless emergency switch.
But for the motor driver that burned up before there was a way to do it.
In the sabertooth there is a pin that if set high it would stop. It won't go anywhere. The sabertooth has an emergency stop function already built in it.

The wireless system is already there it's with an NRF 24L01+ and it's with stm32.
Its pretty simple we already have the pcb that hooks up the NRF.
We already have the a stm32 one that was ordered already. We might have to order another one.
We need to pick up one of those pcbs that where made and handed out to people during embedded 2 for the stm32.

ALWAYS UNPLUG THE JETSON FIRST BEFORE THE MOTOR DRIVERS!!

The NRF modules are kopils so he will take em so we need to get our own. Or just get them from the embedded lab. We might need like 3 to 4 nrfs.
LCD screen.
Sabertooth 2x32
Stm32
3 red boards
Maybe new wires with sockets
Arduino for LED lights if we want it.
Better wire management
Add some sort of ventilation system for the jetson and the boards inside the cabinet.
Main thing we need is the algorithm for the camera and lidar.
GPS they did was just plotting the location and where they where. What we can do is combine Rosas code to the red board plug the gps and it will apply its self as it goes and we can show the gps as a plot on the screen or something.

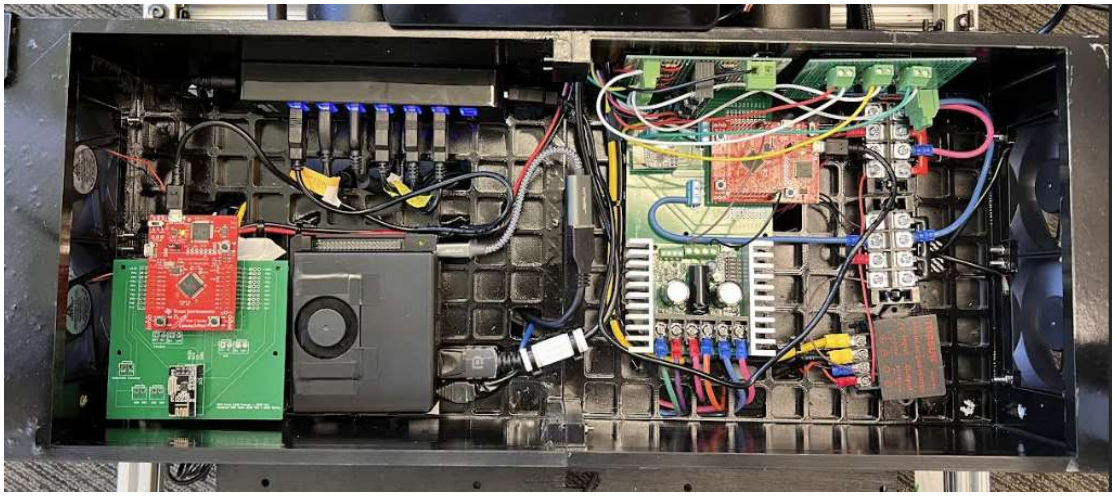
4 nrf to manual control with keyboard. 1 for onboard, 1 for wirless switch, and another nrf for another red board that they have a code to be able to control the car with keyboard. It controls the motors with the arrow keys. NRF are like 7 or 8 bucks.
For the Wireless switch there are 2 for 10$ that has a 1000meter radius the one the last team used were only 100meter.

Might need more white tape.

Would SSH into the jetson to run code
So once everything is connected.
Jetson had issues connecting to Skynet it would change ip address so when they would ssh back in they had to check the ip address again. So having the LCD screen was so easy as all you had to do was look put your mouse there real quick but if you don't have an LCD screen you have to plug in a monitor and check every time for the ip address.

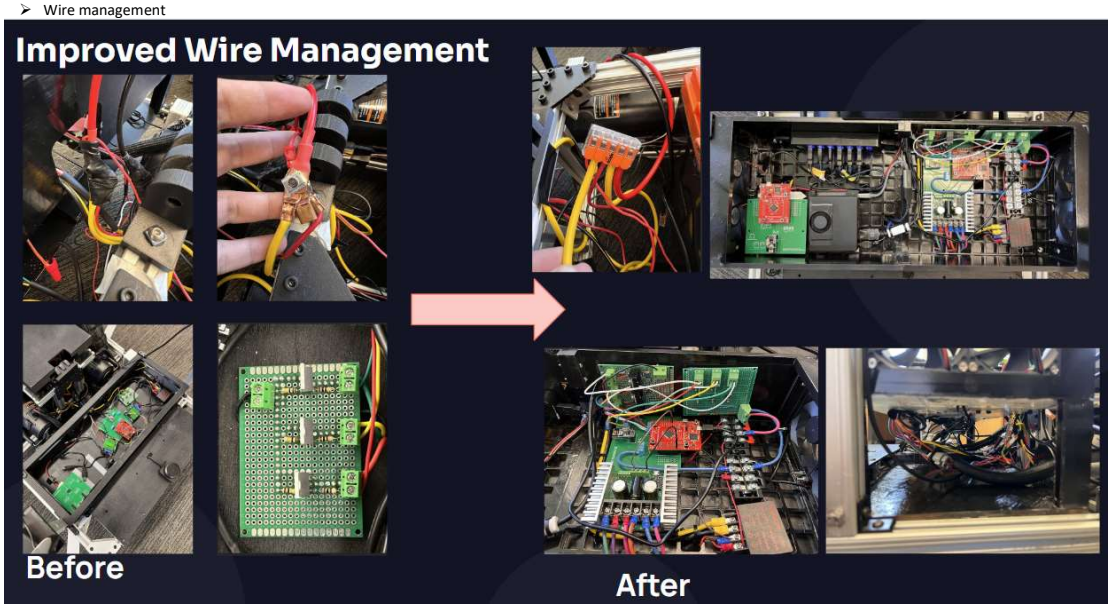
What I didn't add. They talked about something that the video didn't pick up and only my audio did but without visualization to go along it I don't really know what they were talking about.
They talked about replacing 3d printed holes or washers they had to super glue to the wheels. They said something about 5 bolts or 5 holes. Then replacing it with ones they already ordered or made that are made out of metal. They said it was important to do this as in their case they had to tighten up the bolts or screws every so often every time they tested it. Also since they lengths are all different sizes the wheels may have a shimmy that during longer or farther distances can affect the wheels directions that it intends to get from the direction and movement of the motors the wheels are connected to.

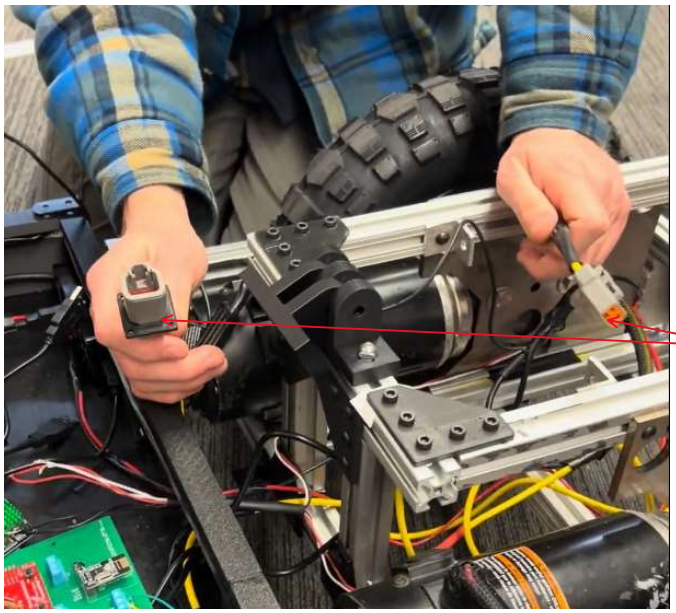
This is what the system looks like now.



The only major changes here are

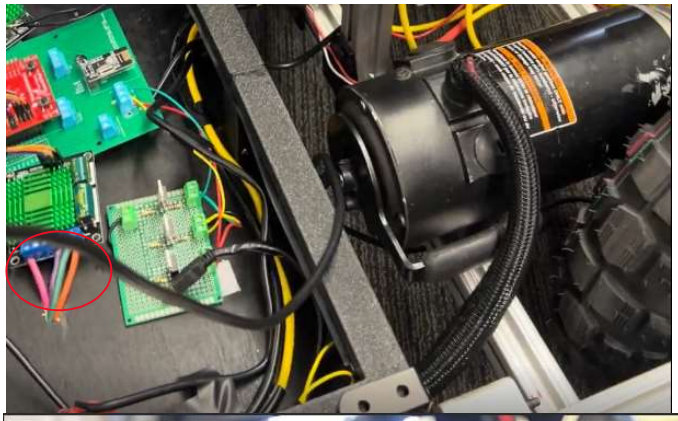
- the encasing itself
  - which I designed to allow ***fans to be installed***. I personally did testing to make sure the airflow inside doesn't escape and flows properly through the exhaust fans.
  - The encasing I designed it to have a ***thin underbelly on the floor*** that houses the hardware. I did this ***so that it is easy to just cut and pop out those thin plastic square flooring in the case so that it can make wire management easier.***
    - To properly cut a square open all you need is a box cutter and easily just press down on the edges of the square thin layer on all 4 edges and then the square will come off.
    - It would be best for you to make sure there are no wires under the square you are trying to cut out so put your hand underneath and find the square with your finger and confirm that the square area is free of wires. If there are wires in the way move them to the side and hold them like that while you cut the square out. Be careful when cutting the edges do it slowly so that you aren't cutting wires or yourself if you have your finger or hand down under.
- Wire management



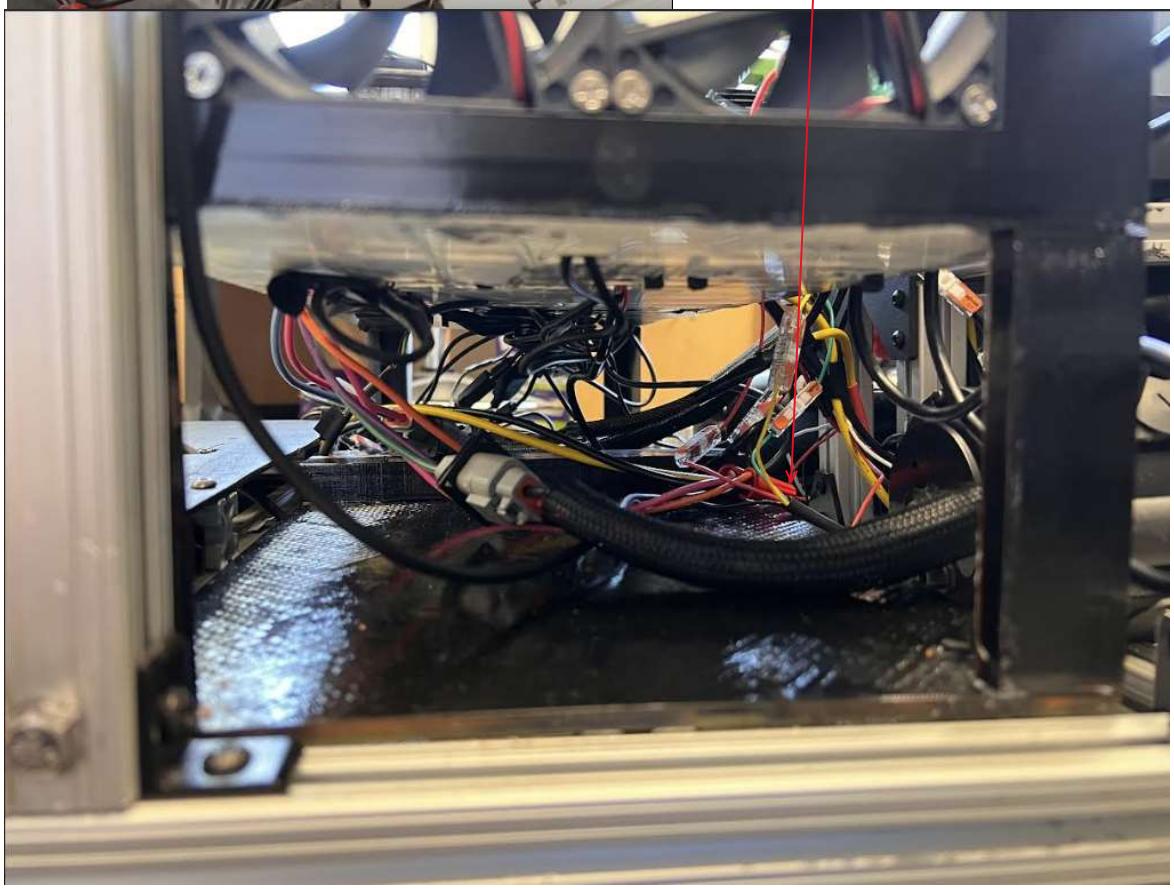


The wiring is the same its just better. As the last team did a rushed wiring job that there was a lot of dead or loose connections that was a pain to debug. So the concepts from section 1 carry over to new setup of the hardware.

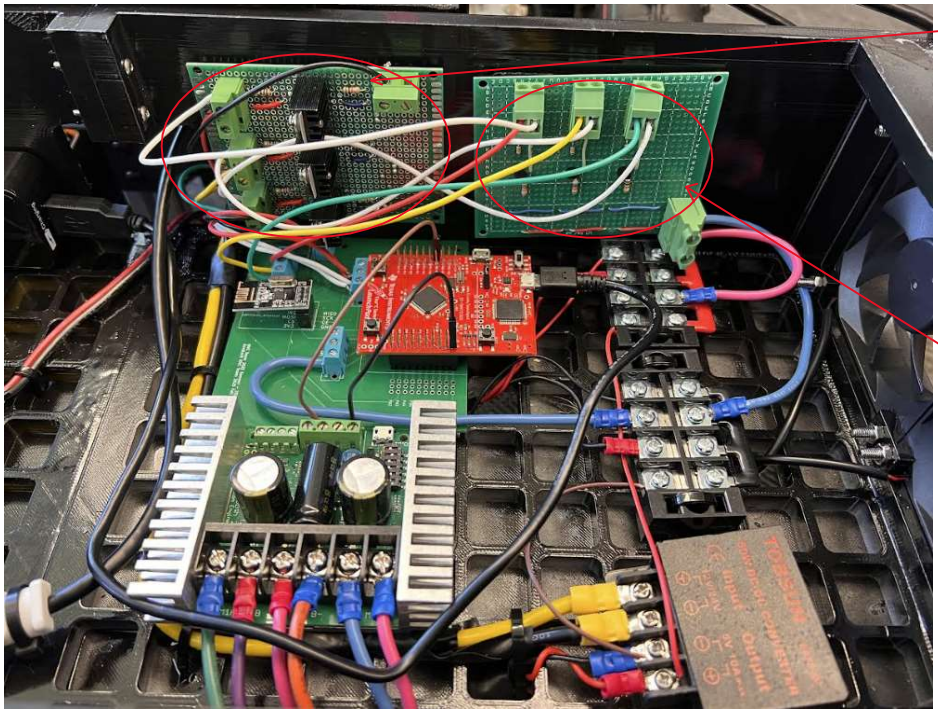
There is no longer any need to worry about connecting these things as we fixed that already by connecting them constantly. The issue before was also solved. About worrying about the Jetson being fried when disconnecting the battery.
But its still best practice when disconnecting and connecting the power/battery make sure to disconnect the jetson itself first.



Look at the hardware yourselves and you will find some of these wire caps that is what Replaced the connections from before. We didn't take a good picture where this is show cased but y'all can see for your selves to get a better understanding of where the wires are going and coming from.

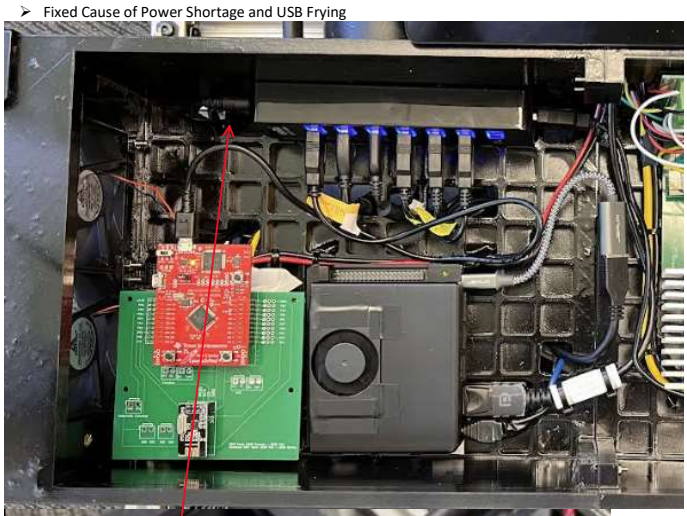


- Improved light circuit

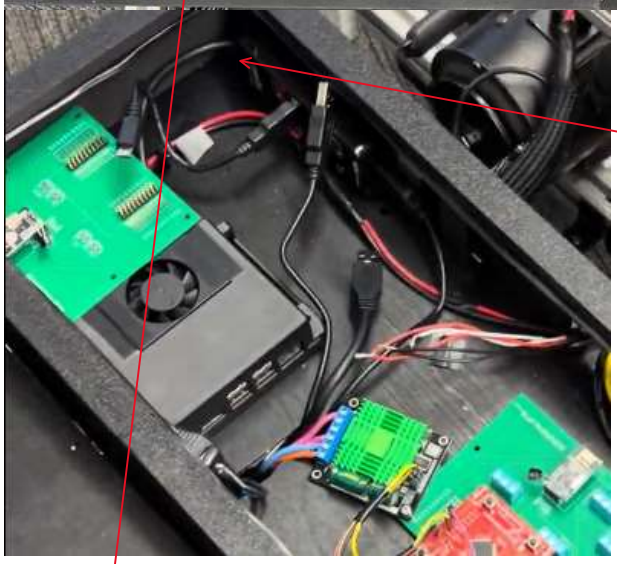


This is the same circuit design as the one shown in section 1 but it
Was improved. The mosfets here are newer and have heat sinks on them.
Though after applying the heat sinks we noticed that the lights were no longer working like before. That is when we found out that the lights were strobe and that the mosfets needed 5V to be able to open and allow the Tiva board to control the power going to the lights.
The main reason why the other mosfet and light combination worked is because the mosfets were overheating which created the gate limit threshold to plummet going from 5V to below to 3.3V which is the amount of voltage the tiva board sends. So instead of getting a command from the tiva board the lights were getting a constant broken stream of 24V which was making the strobe lights act solid which the last team believe was working.
But after adding the new mosfet heatsinks the 5V threshold stayed solid and so the 3.3V provided by the Tiva board no longer was enough to open and control the gate. So I soldered a new circuit that allows the 3.3V to act as a switch and send a 5V charge in the stead of the 3.3 volts the tiva provides, to go to the mosfet connection.

- Fixed Cause of Power Shortage and USB Frying



Current setup



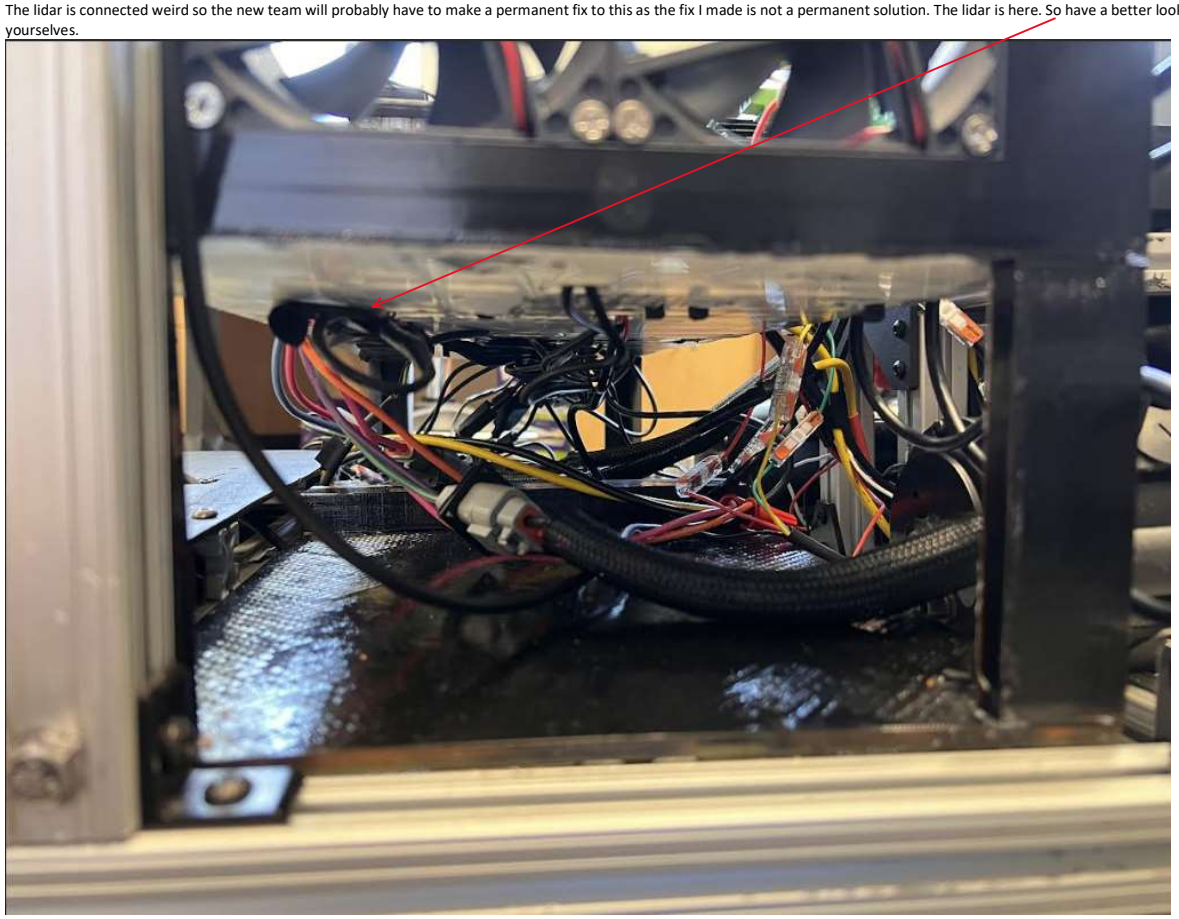
Old teams setup

Nothing is connected here. The charging port was being used to "power" the usb hub which did nothing. So the jetson was the one powering all the ports; the cameras and the Tiva board. Not the lidar though as the lidar had its own power source.

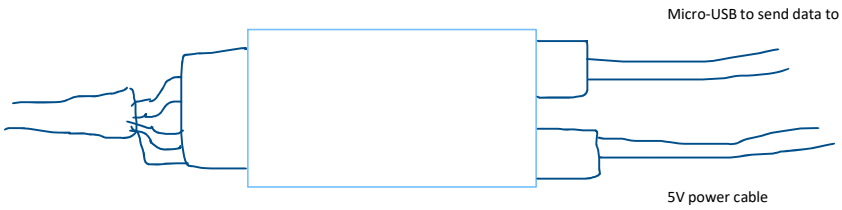
- When I took apart the entire hardware to install the encasing and I put everything back together I found out that the other team had connected the USB hub wrong. Which created too much power to inflow into the jetson which slowly fried all the USB ports on the jetson.
  - Notice the USB hub its an exact expanded model that the last team had. But in ours this cable is connected while the other team didn't have it connected. This is actually the USB hubs power supply. The other team was using the charging port as a power supply for the USB hub they had.
  - The lidar has its own 5v power source and its connected to the usb port. What the other team didn't realize was that the way the lidar is connected led to 5V from the lidar to go into the USB hub. Which caused a problem were if you disconnected the power source (the charging port that the other team though was the power source but wasn't powering anything) of the USB Hub

but kept all the USB ports still connected to the USB hub then the USB hub would still be showing signs of being power on. Our first assumption was that the jetson was powering it but no if you disconnected the USB Hub from the jetson the USB hub was still being powered on. So we didn't know until I debugged it and saw the lidar was the one powering the Hub. Which was the reason why all the jetson's hubs were all fried by the last team. The jetson connection to the USB was receiving power from the lidar and the faulted 5V to the charging port which was only meant to be used as an output to charge things not take in power.

- The lidar is connected weird so the new team will probably have to make a permanent fix to this as the fix I made is not a permanent solution. The lidar is here. So have a better look for yourselves.

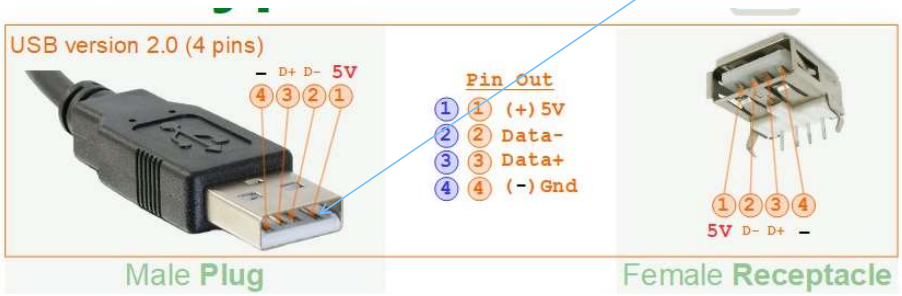


The lidar is connected something like this:



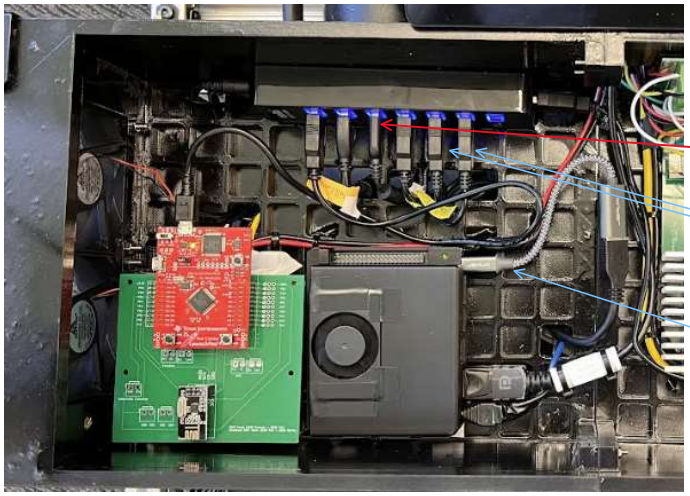
LIDAR connection

My current fix that will need a permanent fix or if not do not unplug the lidar from the USB Hub if not my fix will stop working.
What I did was I added a thin piece of electrical tape and covered the 5V power pin on the usb port. If you were to take out the lidar port then the tape may come off when you try to plug it back in.



If you keep this fix which shouldn't come undone unless you mess with the lidar ports. Its best practice to make sure the USB hub isn't being powered by the lidar to test if its being powered what you need to do is unplug the USB power cable (not the charging cable as that is no longer being used to provide 5V to the Hub but is used to power the receiver TIVA board) and unplug the jetson from the Hub. Then power on the vehicle to the battery and if the Hub lights turn on then the lidar is powering the Hub if its not then the lidar solution I did is still in effect.
If the lights are on do to the lidar powering the hub then you must unplug the lidar and fix the problem either doing what I did or fixing it permanently. The lidar should be the 3rd port from the left.
IF THE LIDAR IS POWERING THE HUB DON'T CONNECT THE HUB TO THE JETSON IT WILL FRY THE LAST REMANINING PORT OR IF YOU HAVE A NEW JETSON IT WILL FRY THAT PORT AND DAMAGE NEAR PORTS ON THE JETSON.

- Discovery of the only working port left

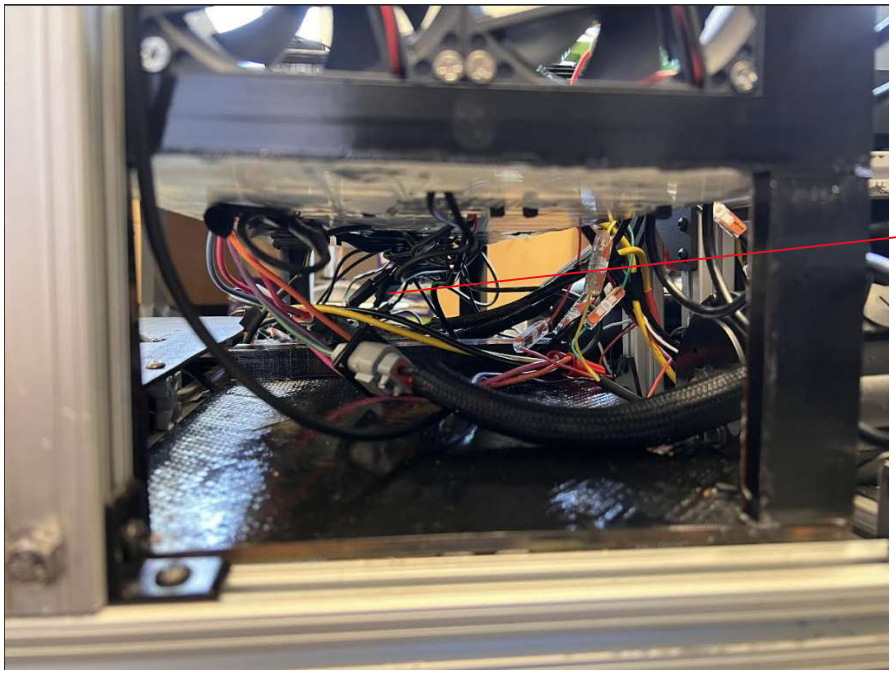


LIDAR: but check the label to make sure it is the lidar. All or most cables are labeled.

The two USB cables to control and power the fan speeds from low to high and on to off.

This is the last remaining port that works on the jetson (at least the Jetson that we were using the professor said he was going to order a new one that would arrive after our semester spring 2026.)
It's a USB-C so you will need a USB C to USB adapter like the one in the image as the one in the image is a personal one we used.

- Changed the USB hub into a 7 port USB hub
- The lights on the vehicle were changed from strobe lights to constant lights
- Added fans



These are the fans connectors and controllers. There are 4 fans and 2 connecters and controller (the connector and controller are a single item) each pair is connected to a connector and controller. The controller has 4 settings off-low-medium-high.
On default we have them on high.

That is all on the new things we added and things that we fixed and debugged.
One thing to note for the next team is to fix the emergency switch under the lights. As currently occasionally when pressed to shut off the vehicle the receiver board also sometimes turns off and then turns back off or stays off. In the past before fixing the connections we burnt 3 Tiva boards because there was back emf going into the board as the board was connected to the 5V power converter and the lights. But after fixing/adding the light circuits and the wiring and putting the Tiva board power cable to the charging port on the USB hub the tiva boards stopped frying. But we noticed this new issue when testing the emergency switch which we didn't really get to debug what it was even doing, why it was doing it, and if it was damaging the board or not as we ran out of time and moved on to finish the autonomous integration code.
So next team will need to fix this issue.